

Module 3 - SQL Analysis Results

Grad Cafe Database Query Analysis

Question 1

How many entries in your database are from applicants who applied for Fall 2026?

Result: Fall 2026 applicant count: 29582

SQL Query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026';
```

Explanation: This query filters the term field to Fall 2026 while ignoring capitalization and extra spaces, then counts every matching row.

Question 2

Among entries that provide a nationality classification, what percentage are international students?

Result: Percent international: 46.29%

SQL Query

```
SELECT
    100.0 * COUNT(*) FILTER (
        WHERE LOWER(TRIM(us_or_international)) = 'international'
    ) / NULLIF(COUNT(*), 0)
FROM applicants
WHERE us_or_international IS NOT NULL
    AND TRIM(us_or_international) <> '';
```

Explanation: The denominator includes only rows with a usable nationality value, while the numerator counts only rows classified as international. NULLIF prevents division by zero.

Question 3

What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Result: Average GPA: 3.85

Average GRE Quantitative: 262.89

Average GRE Verbal: 161.53

Average GRE Analytical Writing: 8.34

SQL Query

```
SELECT
    AVG(gpa),
    AVG(gre),
    AVG(gre_v),
    AVG(gre_aw)
FROM applicants;
```

Explanation: Each AVG is calculated independently. PostgreSQL AVG ignores NULL values, so applicants contribute to each metric only when that specific value is present.

Question 4

What is the average GPA of American applicants who applied for Fall 2026?

Result: Average GPA of American Fall 2026 applicants: 3.79

SQL Query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
    AND LOWER(TRIM(us_or_international)) = 'american'
    AND gpa IS NOT NULL;
```

Explanation: This query restricts rows to Fall 2026 American applicants and includes only records that provide a GPA before calculating the average.

Question 5

What percentage of Fall 2025 entries are acceptances?

Result: Fall 2025 acceptance percentage: 47.92%

SQL Query

```
SELECT
    100.0 * COUNT(*) FILTER (
        WHERE LOWER(TRIM(status)) = 'accepted'
    ) / NULLIF(COUNT(*), 0)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2025';
```

Explanation: The denominator is every Fall 2025 entry, and the filtered count in the numerator includes only accepted entries. This directly produces the requested acceptance percentage.

Question 6

What is the average GPA of accepted applicants who applied for Fall 2026?

Result: Average GPA of accepted Fall 2026 applicants: 3.88

SQL Query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
      AND LOWER(TRIM(status)) = 'accepted'
      AND gpa IS NOT NULL;
```

Explanation: The query limits the data to accepted Fall 2026 applicants and excludes missing GPA values before calculating the average.

Question 7

How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Result: Johns Hopkins master's Computer Science applicant count: 8

SQL Query

```
SELECT COUNT(*)
FROM applicants
WHERE (
    LOWER(program) LIKE '%johns hopkins%'
    OR program ~* '^(^[^a-z])JHU([^a-z]|$)'
)
AND (
    LOWER(program) LIKE '%computer science%'
    OR program ~* '^(^[^a-z])CS([^a-z]|$)'
)
AND (
    LOWER(TRIM(degree)) LIKE 'master%'
    OR LOWER(TRIM(degree)) IN ('ms', 'm.s.')
);
```

Explanation: This query uses the original program and degree fields and allows common variations such as Johns Hopkins/JHU, Computer Science/CS, and master's/MS.

Question 8

How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at Georgetown, MIT, Stanford, or Carnegie Mellon using the original downloaded fields?

Result: Original-field count: 28

SQL Query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
      AND LOWER(TRIM(status)) = 'accepted'
      AND (
          LOWER(TRIM(degree)) IN ('phd', 'ph.d.', 'ph.d')
          OR LOWER(degree) LIKE '%doctor of philosophy%'
      )
      AND (
          LOWER(program) LIKE '%computer science%'
      );
```

```

        OR LOWER(program) LIKE '%comp sci%'
        OR program ~* '^(^a-z)CS(^a-z)|$)'
    )
    AND (
        LOWER(program) LIKE '%georgetown%'
        OR LOWER(program) LIKE '%massachusetts institute of technology%'
        OR program ~* '^(^a-z)MIT(^a-z)|$)'
        OR LOWER(program) LIKE '%stanford%'
        OR LOWER(program) LIKE '%carnegie mellon%'
        OR program ~* '^(^a-z)CMU(^a-z)|$)'
    );

```

Explanation: This query simultaneously filters by term, acceptance status, PhD degree, Computer Science, and the four universities. It also recognizes common abbreviations and naming variations in the original program field.

Question 9

Repeat Question 8 using llm_generated_program and llm_generated_university instead of the original university/program information.

Result: Original-field count: 28

LLM-field count: 28

Difference: 0

SQL Query

```

SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
    AND LOWER(TRIM(status)) = 'accepted'
    AND (
        LOWER(TRIM(degree)) IN ('phd', 'ph.d.', 'ph.d')
        OR LOWER(degree) LIKE '%doctor of philosophy%'
    )
    AND (
        LOWER(llm_generated_program) LIKE '%computer science%'
        OR LOWER(llm_generated_program) LIKE '%comp sci%'
        OR llm_generated_program ~* '^(^a-z)CS(^a-z)|$)'
    )
    AND (
        LOWER(llm_generated_university) LIKE '%georgetown%'
        OR LOWER(llm_generated_university) LIKE '%massachusetts institute of technology%'
        OR llm_generated_university ~* '^(^a-z)MIT(^a-z)|$)'
        OR LOWER(llm_generated_university) LIKE '%stanford%'
        OR LOWER(llm_generated_university) LIKE '%carnegie mellon%'
        OR llm_generated_university ~* '^(^a-z)CMU(^a-z)|$)'
    );

```

Explanation: This repeats the same term, status, and degree restrictions as Question 8 but identifies program and university from the LLM-generated fields. In this dataset the counts are identical, so the LLM-normalized fields did not change the number of matching records for this subset.

Question 10

What percentage of Fall 2026 entries are acceptances?

Result: Fall 2026 acceptance percentage: 37.11%

SQL Query

```
SELECT
    100.0 * COUNT(*) FILTER (
        WHERE LOWER(TRIM(status)) = 'accepted'
    ) / NULLIF(COUNT(*), 0)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026';
```

Explanation: The query uses all Fall 2026 entries as the denominator and counts accepted Fall 2026 entries in the numerator, producing the acceptance percentage for that term.

Question 11

What is the average GPA of international applicants who applied for Fall 2026?

Result: Average GPA of international Fall 2026 applicants: 3.93

SQL Query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
    AND LOWER(TRIM(us_or_international)) = 'international'
    AND gpa IS NOT NULL;
```

Explanation: This query filters to international applicants in Fall 2026 and includes only records with a GPA before calculating the average.